

A multi-objective consistent home healthcare routing and scheduling problem in an uncertain environment

Meng Yang^a, Yaodong Ni^{a,*}, Liu Yang^b

^a School of Information Technology and Management, University of International Business and Economics, Beijing 100029, China

^b Business School, University of International Business and Economics, Beijing 100029, China

ARTICLE INFO

Keywords:

Home health care
Service consistency
Workload balance
Uncertain programming
Multi-objective optimization
Metaheuristics

ABSTRACT

Home health care (HHC) companies serve as the alternative to hospitals aiming to provide customers with medical care at home. A crucial challenge for HHC providers is to optimize routes and schedules for their caregivers to serve customers. Inspired by the practices in the HHC industry, this paper addresses a multi-objective home healthcare routing and scheduling problem (HHRSP) with several conflicting objectives: minimizing routing cost and improving service consistency and workload balance. We refer to the problem as a multi-objective consistent home healthcare routing and scheduling problem (MoConHHRSP). To be more practical, uncertain travel and service times are also considered and defined based on uncertainty theory. Next, the uncertain programming model for the proposed MoConHHRSP is formulated and then reduced to its deterministic equivalent. Due to the NP-hard essence of the problem, an improved multi-objective artificial bee colony (IMOABC) metaheuristic, integrating the large neighborhood search heuristic and an adapted non-dominated solution set update strategy into the multi-objective artificial bee colony (MOABC) framework, is developed. Finally, a series of numerical experiments are conducted to illustrate the competitive performance of the designed algorithm by comparing it with other multi-objective algorithms from multiple evaluation metrics. Furthermore, the trade-off analysis reveals that a better caregiver consistency can be achieved at a high price of total costs and workload balance, while a great improvement on the workload balance can be provided with little deterioration in caregiver consistency. In many cases, low total costs and a high level of workload balance can be achieved simultaneously.

1. Introduction

In response to the development of the aging population and limited medical resources, the home health care (HHC) industry has witnessed a fast-growing in recent years as the alternative of hospitals in many countries and regions (Genet, Boerma, Kroneman, Hutchinson, & Saltman, 2013; Harris-Kojetin, Sengupta, Park-Lee, & Valverde, 2013). HHC provides customers with a series of health care services at home, including illness-care, disabilities-care and elder-care, and so on (Lin, Hung, Liu, & Tsai, 2018; Liu, Yuan, & Jiang, 2017). By assigning caregivers to serve customers at domicile, the congestion of hospitals can be alleviated. In addition, these services are usually less expensive, more convenient and effective than the services delivered in hospitals (Alodhayani, 2017). From another point of view, the preference of people for receiving health care at home also increases (Liu, Yuan, & Jiang, 2019). Inevitably, the HHC industry would continue to develop rapidly owing to the aging population, shortage of medical resources, and preferences of customers.

In HHC, caregivers travel to deliver face-to-face care to customers at their locations. Therefore, home healthcare managers must make valid route plans to assign and route caregivers. Typically, the operations of the HHC include the following procedures (Fikar & Hirsch, 2017; Shi, Boudouh, & Grunder, 2019). Firstly, a set of geographically dispersed customers request multiple services over a certain period. Then the home healthcare managers have to decide the routes for caregivers on each day given the requested services. Finally, the caregivers deliver services for customers according to the planned routes. When making route plans, the home healthcare managers always emphasize the minimization of the total operational costs (Akjiratikar, Yenradee, & Drake, 2007; Liu et al., 2017, 2019; Trautsamwieser & Hirsch, 2014; Yuan, Liu, & Jiang, 2015, 2018). However, customer satisfaction and employee satisfaction also achieve great importance in gaining an advantage in the increasingly competitive HHC industry, as they play a vital role in the profits of HHC firms according to the service profit chain (Sasser, Schlesinger, & Heskett, 1997). Despite the importance of

* Corresponding author.

E-mail address: ni@uibe.edu.cn (Y. Ni).

<https://doi.org/10.1016/j.cie.2021.107560>

customer satisfaction and caregiver satisfaction, the related research is still in scarcity, which motivates us to introduce a multi-objective home healthcare routing and scheduling problem (HHRSP) to reconcile the interests of the different stakeholders in the HHC.

As mentioned in Woodward, Abelson, Tedford, and Hutchison (2004), in home care, customers are very sensitive to the change of the service providers. Clients who are cared for with high consistency of service providers, that is service provided by the same caregiver or as few different caregivers as possible, express the greatest satisfaction with service provision. The research of Russell, Rosati, Rosenfeld, and Marren (2011) also shows that higher levels of consistency of care lead to high patient satisfaction. The benefits of personnel consistency are twofold. On one hand, the staff can become familiar with customer needs and requirements and provide a better-customized service (Janssens, Van den Bergh, Sörensen, & Cattrysse, 2015; Rodriguezmartin, Salazargonzalez, & Yaman, 2019). On the other hand, customers can also be comfortable with the service provider if they are always served by the same one or a familiar one, especially in home-care services when physical contact is required (Paquette, Bellavance, Cordeau, & Laporte, 2012; Woodward et al., 2004). Time consistency which requires services to be delivered at the almost same time is also very important because it helps the customers better plan and organize their time (Duque, Castro, Sörensen, & Goos, 2015). As a consequence, improving service consistency including caregiver consistency and time consistency is a good way to facilitate a high level of customer satisfaction. Besides, workload balance can ensure the equity of caregivers and increase employee satisfaction (Matl, Hartl, & Vidal, 2018), which directly works on the quality of service and customer satisfaction and thus on the profits of HHC companies. Taking these concerns into account, this paper initiates a multi-objective consistent home healthcare routing and scheduling problem (MoConHHRSP), in which routing cost, caregiver consistency, and workload balance are independent objectives while time consistency is enforced by constraints.

Additionally, previous research mainly focuses on the deterministic model assuming travel and service times to be exact and known. However, in practice, travel and service times often involve various types of uncertainties. Varying road and traffic conditions, such as rush hours and congestion, practical weather conditions, and driving states of caregivers often lead to uncertain travel times. The service time of each customer is related to his physical condition, which cannot be determined before the service is completed, especially sometimes emergencies are involved. Thus, if the caregivers follow the routes made in deterministic models, the uncertainties in travel and service times may result in late arrival at customers. The delayed service for the current customer and the delayed or even failed service for the subsequent customers in the route will reduce customer satisfaction and impact the reputations of HHC companies negatively. Therefore, it is of great significance to consider the robustness of the uncertain travel and service times when making route plans in HHC.

In conclusion, we propose a multi-objective consistent HHRSP considering workload balance in an uncertain environment. The main contributions of our work are summarized as follows:

- A new home healthcare optimization model with considerations of service consistency and workload balance under an uncertain environment is developed and modeled as a multi-objective problem.
- Uncertainty theory is utilized to define uncertain travel and service times and the uncertain programming model is formulated and then transformed into its deterministic equivalent.
- An improved multi-objective artificial bee colony (IMOABC) heuristic, embedding adapted non-dominated solution set update strategy and large neighborhood search (LNS) into multi-objective artificial bee colony (MOABC) framework, is designed.

The remainder of this paper is organized as follows. Section 2 overviews the related literature to the proposed MoConHHRSP. The problem description and model formulation are presented in Section 3. A multi-objective solution framework is developed in Section 4. Computational experiments are conducted and the analysis is given in Section 5. Section 6 summarizes this paper with some conclusions and future research directions.

2. Related literature

The HHRSP has attracted extensive interest from scholars all over the world in the past two decades (Cissé, et al., 2017; Fikar & Hirsch, 2017). Despite the importance of consistency and workload balance on customer satisfaction in the HHRSP, few papers focus on these features. In our literature review, we survey the studies concerning the following aspects: the HHRSP with workload balance, service consistency, multi-objective optimization, and uncertainties in travel and service times.

2.1. HHRSP with workload balance

With the rapid growth of HHC services, more and more researchers have studied issues involving the routing and scheduling of home health caregivers. Begur, Miller, and Weaver (1997) are among the earliest to study the HHRSP, in which a spatial decision support system is developed to address a home healthcare routing problem in the United States. Later, Akjiratikarl et al. (2007) model the home healthcare scheduling problem as the VRP, and the particle swarm optimization algorithm is applied to the proposed problem. Motivated by the practices in HHC, workload balance has also been introduced in the optimization models of the HHRSP. Reasonably, workload balance contributes to the fairness of employees and facilitates employee satisfaction (Milburn & Spicer, 2013), which plays a significant role in the service quality when they interact with customers. Besides, by balancing the workload of employees, the route plans can also be more valid because the workload balance can help reduce the failure of service delivery and reduce overtime (Matl et al., 2018; Matl, Hartl, & Vidal, 2019). As mentioned in Matl et al. (2019), there is no uniform measure method of workload balance. In the literature, the workload is measured by different metrics and the workload balance is defined by different functions. Liu, Xie, and Garaix (2013) consider workload balance as minimizing the maximum route duration. Milburn and Spicer (2013) define the workload as the number of assigned clients for each nurse over the planning horizon and strives to minimize the number of all pairwise differences between nurses of the workload. Decerle, Grunder, El Hassani, and Barakat (2019) define the workload as the working time of a caregiver and optimize the maximal working time difference among caregivers.

2.2. Service consistency

Service consistency is first introduced by Groer, Golden, and Wasil (2009), requiring that customers are always visited by the same driver (personnel consistency) at the almost same time (time consistency) each day when they ask for services over a planning horizon. Following the definition of Groer et al. (2009), service consistency has been highlighted in the VRP involving various application areas, including small package delivery (Kovacs, Parragh, & Hartl, 2014; Zhen, Lv, Wang, Ma, & Xu, 2020), transportation of the disabled and the elderly (Feillet, Garaix, Lehuède, Peton, & Quadri, 2014), cleaning services (Tarantilis, Stavropoulou, & Repoussis, 2012), pharmaceutical distribution (Campelo, Nevesmoreira, Amorim, & Almadalobo, 2019). As far as personnel consistency is concerned, some research has relaxed it to a limited number of different drivers serving per customer during the given period and modeled it as the constraints (Braekers & Kovacs, 2016; Kovacs, Golden, Hartl, & Parragh, 2015; Luo, Qin, Che, & Lim, 2015) or the objective (Kovacs, Parragh, & Hartl, 2015; Lian, Milburn,

& Rardin, 2016). A new measure of personnel consistency proposed by Yang, Ni, and Song (2020) quantifies personnel consistency based on information entropy and can be modeled as an objective function. Time consistency is often measured by the maximum difference among arrival times of each day for each customer (Campelo et al., 2019; Groer et al., 2009; Zhen et al., 2020). Other researchers focus on time consistency through the same time window (Song, Ulmer, Thomas, & Wallace, 2020; Spliet, Dabia, & Van Woensel, 2018; Spliet & Gabor, 2015). Our paper differs from the above papers for the following reasons. Caregiver consistency is modeled as objective and defined by the method of Yang et al. (2020), and time consistency is enforced as the soft constraints by the same time point on each day, which is common in the home service routing problem (Zhan, Wang, & Wan, 2021). Moreover, we consider service consistency in the context of HHC thus service consistency is tackled together with other aspects, such as workload balance, overtime cost.

2.3. Multi-objective optimization of the HHRSP

The recent advance in the HHC research has moved to explore the HHRSP with a multi-objective optimization method to obtain the Pareto frontier instead of a weighted objective function (Akjiratikarl et al., 2007; Duque et al., 2015; Milburn & Spicer, 2013; Trautsamwieser & Hirsch, 2014). The bi-objective model in HHRSP is initiated by Braekers, Hartl, Parragh, and Tricoire (2016), which focuses on two objectives: minimize the total cost including routing and overtime costs and minimize client inconvenience measured by the deviation from the preferred visit time of the assigned nurses. A metaheuristic based on local searches and dynamic programming is proposed to achieve this goal. Recently, as customer satisfaction is playing an increasingly important role in the performance of the companies, patients' satisfaction is also incorporated into the research of HHRSP as an objective (Fathollahi-Fard, Ahmadi, Goodarzi, & Cheikhrouhou, 2020; Liu, Yang, Su, & Xu, 2018). Patients' satisfaction is characterized as the priority of the patients to the caregivers assigned to provide services for them (Fathollahi-Fard et al., 2020) and the total additional service length (Liu et al., 2018), respectively. Since obtaining non-dominated solutions in reasonable computation time for the multi-objective HHRSP is often infeasible, heuristic approaches are often utilized to approximate the Pareto frontier (Goodarzi, Tavakkoli-Moghaddam, & Amini, 2020; Xu, Hao, & Zheng, 2020). The research on multi-objective optimization for the HHRSP deriving a set of Pareto optimal solutions is yet not common in HHRSP. Based on this finding, we initiate a multi-objective optimization model by considering several objective functions: minimizing routing cost, improving caregiver consistency and workload balance. Moreover, a multi-objective approach is developed to obtain different solutions so that the decision-makers of HHC can choose to balance multiple conflicting objectives.

2.4. Uncertainties in travel and service times

Although extensive papers have been conducted on the HHRSP from various perspectives, only a few of them take the uncertain travel and service times in the HHRSP into account. Stochastic service times have been studied in home service routing with routing and appointment scheduling decisions integrated (Zhan & Wan, 2018; Zhan et al., 2021). Yuan et al. (2015) consider stochastic service times and skill requirements simultaneously. A stochastic programming model is defined with tardiness at customers penalized in the total routing cost. Stochastic service times and travel times are investigated in Yuan et al. (2018), in which a stochastic programming model with recourse is proposed and penalties of skipping customers with violated time windows are added to the total cost. The stochasticity of travel times and service times is also addressed in Liu et al. (2019), while the difference lies in the context of multi-depot. Shi et al. (2019) propose a robust optimization model for the HHRSP with uncertain travel and service

times. From the above papers, we can see that existing research has never incorporated travel and service time uncertainty in the HHRSP with the considerations of service consistency and workload balance. Therefore, we focus on investigating the uncertain travel and service times in the HHRSP with the considerations of service consistency and workload balance. As the historical data is often inferior to estimate these parameters in real life (Shi et al., 2019), uncertainty theory is utilized to deal with parameters that lack statistical data (Liu, 2009).

By reviewing the related papers, the findings can be summarized as follows:

- a majority of work only considers the optimization of operating costs while ignoring customer satisfaction, especially, service consistency has not been incorporated into the HHRSP.
- most of the studies focus on the deterministic model without considering practical uncertainties, uncertain models are still scarce in the literature.
- The existing research prefers to formulating the HHRSP as a single objective optimization problem or a weighted objective function, the multi-objective home healthcare models are still in their infancy.

Therefore, in this paper, we study the MoConHHRSP under an uncertain environment. The problem is defined as a multi-objective, multi-period, and single-depot home routing optimization problem under the constraint of a maximum working hour of caregivers. Travel times and service times are defined as uncertain variables and tackled by uncertainty theory (Liu, 2007, 2010). The uncertain programming model of the proposed problem is formulated and then the deterministic equivalent is proposed. Considering the essence of NP-hardness of routing and scheduling problems, an IMOABC is developed with improvements in the non-dominated solution set update strategy and neighborhood search. Computational experiments are conducted to validate the performance of the designed IMOABC by comparing it with the MOABC algorithm and the popular non-dominated sorting genetic algorithm-II (NSGAII). The trade-off analysis is also provided.

3. Problem description and model formulation

3.1. Problem description

In this subsection, the uncertain MoConHHRSP is described as follows. During a multi-day planning horizon, there is a set of geographically dispersed customers who may require single-day or multi-day services. The subset of customers who ask for service every day can be determined by the booking information provided by customers. Therefore, on each day, the locations and the service dates are predefined by customers. The customers also designate their allowable service start times when they request services. By fixing the allowable service start time of each customer over the given period, time consistency can be imposed because customers are serviced almost at the same time point on each day when they have service requirements.

A set of caregivers are responsible to travel to deliver services to customers. The company organizes a home health care center where all caregivers start their work and return after serving the assigned customers. Generally, caregivers drive to the locations of customers, generating the travel cost related to their travel times. The travel time between two locations is often uncertain due to the varying practical conditions. Besides, the care service time of each customer is uncertain. As a consequence, caregivers may arrive at customers earlier or later than their specified times. In most applications, one customer would not permit a service if the caregiver arrives at the customer earlier. The caregiver only can wait until the allowable service start time. However, if caregivers serve the customers later than their predefined times, the penalty of caregivers is incurred as late arrival will lead to dissatisfaction. The unit penalty cost is determined by the company. Caregivers have a maximum working time each day. Workload balance

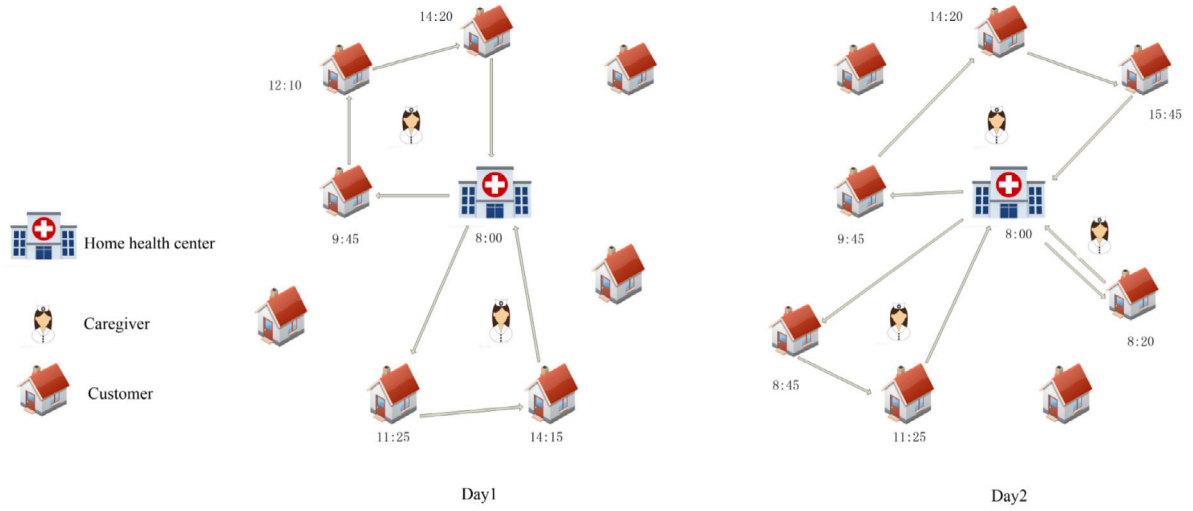


Fig. 1. An example of the proposed MoConHHRSP.

and caregiver consistency in the given period are taken into account in the proposed problem. This problem aims to construct daily routes for a set of caregivers over a certain planning horizon. A simple example is illustrated in Fig. 1. This figure shows eight customers, three caregivers in a planning horizon of two days. The daily routes of each caregiver for visiting customers in a working day are presented. We can see that some customers may require to be served for two days and some customers only have a single-day service requirement, and their specified service start times are the same over the two days. The example routes achieve the best caregiver consistency for each customer.

3.2. Model formulation

In this section, the multi-objective mathematical model for the proposed consistent home healthcare routing and scheduling problem with uncertain travel and service times is presented. The MoConHHRSP is defined on a complete connected graph $G = (V, E)$, where $V = \{0, N\}$ is the nodes set and $E = \{(i, j) | i, j \in V, i \neq j\}$ is the set of edges. Node 0 denotes the home healthcare center and $N = \{1, 2, \dots, |N|\}$ represents the set of customers. This problem is described in the context of multiple days. Let $D = \{1, 2, \dots, |D|\}$ represent the service duration. A set of caregivers K is available to provide service for customers. The assumptions of the proposed home healthcare optimization model are classified as follows:

- (1) Each customer can be visited at most once a day.
- (2) Each customer has the same specified service start time on each day.
- (3) Caregivers are homogeneous and available at the home healthcare center.
- (4) Each caregiver starts and ends at the home healthcare center on each day.
- (5) Each caregiver can operate for a maximum working time T on each day.
- (6) The departure time from the home healthcare center is 0.
- (7) Travel and service times are uncertain.

The sets, indices, parameters, and decision variables of the proposed model are introduced as follows:

Sets and Indices:

- V Set of nodes, $V = \{0, N\}$ and 0 denotes the health care center.
- N Set of customers.
- K Set of caregivers.
- D Set of service days.
- i, j Index of node (customer or health center).
- k, u Index of caregiver.
- d Index of day.

Parameters:

- T The maximum working time of each caregiver.
- g_i The sum of the days that customer i needs service in the planning horizon.
- ct Unit travel time cost
- cs Unit service time cost
- cp Unit penalty cost of being late
- h_k The workload of caregiver k
- l_i The specified service start time of customer i
- w_{id} Equals 1 if customer i asks for service on day d , otherwise 0.
- $\tilde{t}_{ijd}$ Travel time between node i and j on day d .
- $\tilde{s}_{id}$ Service time at customer i on day d .

Decision variables:

- x_{ijkd} Binary variable, equals 1 if customer j is serviced after customer i by caregiver k on day d and equals 0 otherwise.
- y_{jkd} Binary variable, equals 1 if customer j is visited by caregiver k on day d and equals 0 otherwise.
- z_{ik} Binary variable, equals 1 if customer i is assigned to caregiver k and equals 0 otherwise.
- v_i The number of caregivers assigned to customer i over the planning horizon.
- b_{ik} The number of days of customer i served by caregiver k .
- r_{id} Actual service start time at customer i on day d , equals 0 if customer i asks no service on day d .

3.3. Uncertain programming model for the MoConHHRSP

In our proposed problem, travel and service times are uncertain variables. They are involved in the objective of minimizing total routing costs and some constraints. However, an objective function involving uncertain variables cannot be directly optimized and uncertain constraints also do not define a crisp feasible set. Therefore, we use uncertain programming proposed by Liu (2009) to formulate the uncertain programming model of the proposed problem. The value of an uncertain objective function is estimated by its expected value and the uncertain constraints hold with confidence levels. The uncertain programming model is defined as follows:

$$(M0) \quad f_1 = \min \mathbb{E} \left[ct \sum_{d \in D} \sum_{k \in K} \sum_{i \in V} \sum_{j \in V, j \neq i} x_{ijkd} \tilde{t}_{ijd} + cs \sum_{d \in D} \sum_{k \in K} \sum_{j \in N} y_{jkd} \tilde{s}_{jd} + cp \sum_{d \in D} \sum_{j \in N} (r_{jd} - l_j) \right], \quad (1)$$

$$f_2 = \min \sum_{i \in N} \begin{cases} v_i + \frac{1}{\log_2 v_i} (-\sum_{k \in K} \frac{b_{ik}}{g_i} \log_2 \frac{b_{ik}}{g_i}) & v_i \neq 1 \\ 1 & v_i = 1, \end{cases} \quad (2)$$

$$f_3 = \min \sum_{k \in K} \sum_{u \in K: u \geq k} |h_k - h_u|, \quad (3)$$

s.t.

$$r_{0d} = 0, \quad \forall d \in D, \quad (4)$$

$$y_{0kd} = 1, \quad \forall k \in K, d \in D, \quad (5)$$

$$\sum_{k \in K} y_{ikd} = w_{id}, \quad \forall i \in N, d \in D, \quad (6)$$

$$\sum_{i \in V, i \neq j} x_{ijkd} = \sum_{i \in V, i \neq j} x_{jikd} = y_{jkd}, \quad \forall j \in N, k \in K, d \in D, \quad (7)$$

$$\sum_{d \in D} y_{ikd} = b_{ik}, \quad \forall i \in N, k \in K, \quad (8)$$

$$\sum_{k \in K} b_{ik} = g_i, \quad \forall i \in N, \quad (9)$$

$$\sum_{k \in K} z_{ik} = v_i, \quad \forall i \in N, \quad (10)$$

$$z_{ik} \geq y_{ikd}, \quad \forall i \in N, k \in K, d \in D, \quad (11)$$

$$h_k = \sum_{d \in D} \sum_{i \in N} y_{ikd}, \quad \forall k \in K, \quad (12)$$

$$\sum_{i, j \in S, i \neq j} x_{ijkd} \leq |S| - 1, \quad \forall S \subseteq N, 2 \leq |S| \leq |N|, k \in K, d \in D, \quad (13)$$

$$\mathcal{M}\{r_{jd} + \tilde{s}_{jd} + \tilde{t}_{j0d} \leq T\} \geq \beta_{jd}, \quad \forall j \in N, d \in D, \quad (14)$$

$$x_{ijkd} \in \{0, 1\}, \quad \forall i \in N, j \in N, j \neq i, k \in K, d \in D, \quad (15)$$

$$y_{jkd} \in \{0, 1\}, \quad \forall j \in N, k \in K, d \in D, \quad (16)$$

$$z_{ik} \in \{0, 1\}, \quad \forall i \in N, k \in K, \quad (17)$$

$$r_{id} \geq 0, \quad \forall i \in N, d \in D. \quad (18)$$

$$b_{ik} \geq 0 \text{ and is an integer}, \quad \forall i \in N, k \in K, \quad (19)$$

$$v_i \geq 0 \text{ and is an integer}, \quad \forall i \in N. \quad (20)$$

The objective function (1) aims to minimize the total operating cost for all customers over the given planning horizon, measured by the sum of travel cost, service cost, and penalty of late arrival at customers. The first term is the travel cost, the second is the travel cost and the last one is the penalty of delayed service for customers. The objective function (2) minimizes the value of caregiver inconsistency to maximize caregiver consistency, measured by the service length ratio of caregivers over the planning horizon and the formulation evolves from the formula of information entropy. The objective function (3) aims at improving workload balance by minimizing workload imbalance, measured by the sum of all pairwise differences between caregivers of the number of assigned customer service requests over the planning horizon.

Constraints (4) and Constraints (5) ensure that each caregiver starts working from the depot at time 0. Each customer is visited once on each day service request is required in Constraints (6). Equalities (7) are flow conservation constraints at each node. Constraints (8) and (9) define the service days of each caregiver per customer. The calculation of the number of caregivers for each customer is given in Constraints (10). Constraints (11) indicate that if one caregiver provides service for one customer in the planning period, then the caregiver is assigned to the customer. Constraints (12) determine the workload of each caregiver. Constraints (13) are the subtour elimination constraints. In Constraints (14), β_{jd} denotes the predefined confidence level that the uncertain constraints hold. Constraints (14) guarantee that the working time of each caregiver cannot exceed the maximum working time with the confidence level of β_{jd} . Thus, a higher belief degree β_{jd} means a higher measurement of overwork of caregivers. Constraint (15)–(20) define the domains of decision variables.

3.4. Calculating the actual service start times

In model (M0), the calculation of the actual service start time r_{jd} at a customer $j \in N$ on day $d \in D$ needs further elaboration. Due to uncertain travel and service time, the actual service start time at the customer r_{jd} is also uncertain. In this section, based on uncertainty theory, we give the method of calculating the actual service start times using inverse uncertainty distributions of uncertain variables.

Firstly, the actual service start time r_{jd} at customer $j \in N$ on day $d \in D$ should satisfy that the caregiver has arrived at the customer (the actual arrival time of the caregiver) and the customer is ready to be served (the specified service start time of the customer). We introduce a_{jd} to denote the actual arrival time at customer $j \in N$ on day $d \in D$. Thus we can get

$$r_{jd} = \max\{a_{jd}, l_j\}.$$

The uncertain variable a_{jd} is independent of the constant l_j . Assume r_{jd} and a_{jd} have regular uncertainty distribution $\Phi_{r_{jd}}$ and $\Psi_{a_{jd}}$, according to the operational laws for calculating the inverse uncertainty distributions of strictly increasing function of uncertain variables in uncertainty theory (Details can be referred to the appendix), we get

$$\Phi_{r_{jd}}^{-1}(\alpha) = \Psi_{a_{jd}}^{-1}(\alpha) \vee l_j,$$

for all $j \in N, d \in D$, where α is the confidence level. Clearly, this equality can be further relaxed to the following inequalities:

$$\Phi_{r_{jd}}^{-1}(\alpha) \geq l_j,$$

$$\Phi_{r_{jd}}^{-1}(\alpha) \geq \Psi_{a_{jd}}^{-1}(\alpha),$$

for all $j \in N, d \in D$, where α is the confidence level. Next, we give the lemma and the proof to show how to obtain $\Psi_{a_{jd}}^{-1}(\alpha)$.

Lemma 1. Assume that $\tilde{t}_{ijd}$ and $\tilde{s}_{id}$ are independent uncertain variables with regular uncertainty distributions $\Phi_{\tilde{t}_{ijd}}$ and $\Phi_{\tilde{s}_{id}}$, respectively. Then the arrival time a_{jd} is an uncertain variable whose inverse uncertainty distribution is

$$\Psi_{a_{jd}}^{-1}(\alpha) = \sum_{i \in N, i \neq j} \sum_{k \in K} x_{ijkd} (\Phi_{r_{id}}^{-1}(\alpha) + \Phi_{\tilde{s}_{id}}^{-1}(\alpha) + \Phi_{\tilde{t}_{ijd}}^{-1}(\alpha)), \quad \forall j \in N, d \in D,$$

where α is the predefined confidence level.

Proof. a_{jd} is the arrival time at one customer $j \in N$ on one day $d \in D$. We have noted that all caregivers start from the home health center at time 0 every day, then $r_{0d} = 0$ and $s_{0d} = 0$. As each customer has exactly one predecessor, for each $j \in N$ there exists exactly one $i \in V, i \neq j$ that makes $x_{ijkd} = 1$, and $\sum_{i \in V, i \neq j} x_{ijkd} = 1$, which denotes that customer j is visited after node i . Then the arrival time at customer j equals the sum of the actual service start time at node i plus the service time at node i plus the travel time between i and j , that is $a_{jd} = r_{id} + \tilde{s}_{id} + \tilde{t}_{ijd}$, thus the arrival time a_{jd} can be obtained from the following equation

$$a_{jd} = \sum_{i \in V, i \neq j} \sum_{k \in K} x_{ijkd} (r_{id} + \tilde{s}_{id} + \tilde{t}_{ijd}), \quad \forall j \in N, d \in D.$$

This recursive process can produce all arrival times for customers.

Assume that $\tilde{t}_{ijd}$ and $\tilde{s}_{id}$ are independent uncertain variables with regular uncertainty distributions $\Phi_{\tilde{t}_{ijd}}$ and $\Phi_{\tilde{s}_{id}}$, respectively. a_{jd} is increasing with uncertain variable r_{id} , $\tilde{t}_{ijd}$ and $\tilde{s}_{id}$. Based on the operational law for calculating the inverse uncertainty distributions of strictly increasing function of uncertain variables in uncertainty theory (Details can be referred to the appendix), the arrival time a_{jd} is an uncertain variable whose inverse uncertainty distribution is

$$\Psi_{a_{jd}}^{-1}(\alpha) = \sum_{i \in N, i \neq j} \sum_{k \in K} x_{ijkd} (\Phi_{r_{id}}^{-1}(\alpha) + \Phi_{\tilde{s}_{id}}^{-1}(\alpha) + \Phi_{\tilde{t}_{ijd}}^{-1}(\alpha)), \quad \forall j \in N, d \in D,$$

where α is the predefined confidence level. Thus the lemma is proved.

3.5. Deterministic equivalent of the proposed model

To solve (M0), the expected values and uncertain measures of the uncertain variables need to be computed, which is complicated and time-consuming. Therefore, in this section, taking advantage of some properties of uncertainty theory (Liu, 2007), we transform model (M0) into its deterministic equivalent.

Theorem 1. Model (M0) is equivalent to the following crisp mathematical programming model:

$$(M1) \quad f_1 = \min \left[ct \sum_{d \in D} \sum_{k \in K} \sum_{i \in V} \sum_{j \in V, j \neq i} x_{ijkd} \int_0^1 \Phi_{t_{ijkd}}^{-1}(\alpha) d\alpha \right. \\ \left. + cs \sum_{d \in D} \sum_{k \in K} \sum_{j \in N} y_{jk d} \int_0^1 \Phi_{s_{jd}}^{-1}(\alpha) d\alpha \right. \\ \left. + cp \sum_{d \in D} \sum_{j \in N} \int_0^1 \Phi_{r_{jd}}^{-1}(\alpha) d\alpha - l_j \right], \quad (21)$$

$$f_2 = \min \sum_{i \in N} \begin{cases} v_i + \frac{1}{\log_2 v_i} (-\sum_{k \in K} \frac{b_{ik}}{g_i} \log_2 \frac{b_{ik}}{g_i}) & v_i \neq 1 \\ 1 & v_i = 1, \end{cases} \quad (22)$$

$$f_3 = \min \sum_{k \in K} \sum_{u \in K: u \geq k} |h_k - h_u|, \quad (23)$$

s.t.

$$(4)-(13), \quad (24)$$

$$\Phi_{r_{jd}}^{-1}(\beta_{jd}) \geq l_j, \quad \forall j \in N, d \in D, \quad (25)$$

$$\Phi_{r_{jd}}^{-1}(\beta_{jd}) \geq \Psi_{a_{jd}}^{-1}(\beta_{jd}), \quad \forall j \in N, d \in D, \quad (26)$$

$$\Phi_{r_{jd}}^{-1}(\beta_{jd}) + \Phi_{s_{jd}}^{-1}(\beta_{jd}) + \Phi_{t_{j0d}}^{-1}(\beta_{jd}) \leq T, \quad \forall j \in N, d \in D, \quad (27)$$

$$\Psi_{a_{jd}}^{-1}(\beta_{jd}) = \sum_{i \in N, i \neq j} \sum_{k \in K} x_{ijkd} (\Phi_{a_{id}}^{-1}(\beta_{jd}) + \Phi_{s_{id}}^{-1}(\beta_{jd}) + \Phi_{t_{ij d}}^{-1}(\beta_{jd})), \\ \forall j \in N, k \in K, d \in D, \quad (28)$$

$$(15)-(20). \quad (29)$$

Proof. We first deal with the uncertain objective function (1). The objective function f_1 is strictly increasing with respect to $\tilde{t}_{ijkd}$, $\tilde{s}_{jd}$ and r_{jd} . Assume the travel time $\tilde{t}_{ijkd}$ and the service time $\tilde{s}_{jd}$ are uncertain variables with regular uncertainty distribution $\Phi_{t_{ijkd}}$ and $\Phi_{s_{jd}}$, according to the conclusion in the above section, then r_{jd} has an inverse uncertainty distribution

$$\Phi_{r_{jd}}^{-1}(\alpha) \geq l_j,$$

$$\Phi_{r_{jd}}^{-1}(\alpha) \geq \Psi_{a_{jd}}^{-1}(\alpha),$$

where

$$\Psi_{a_{jd}}^{-1}(\alpha) = \sum_{i \in N, i \neq j} \sum_{k \in K} x_{ijkd} (\Phi_{a_{id}}^{-1}(\alpha) + \Phi_{s_{id}}^{-1}(\alpha) + \Phi_{t_{ij d}}^{-1}(\alpha)).$$

Then we can compute the expected value of the total costs and the objective function (1) is equal to

$$f_1 = \min \left[ct \sum_{d \in D} \sum_{k \in K} \sum_{i \in V} \sum_{j \in V} x_{ijkd} \int_0^1 \Phi_{t_{ijkd}}^{-1}(\alpha) d\alpha \right. \\ \left. + cs \sum_{d \in D} \sum_{k \in K} \sum_{j \in N} y_{jk d} \int_0^1 \Phi_{s_{jd}}^{-1}(\alpha) d\alpha \right. \\ \left. + cp \sum_{d \in D} \sum_{j \in N} \int_0^1 \Phi_{r_{jd}}^{-1}(\alpha) d\alpha - l_j \right]. \quad (30)$$

Next, we deal with the uncertain Constraints (14). Each constraint function in Constraints (14) is strictly increasing at $\tilde{t}_{ijkd}$, $\tilde{s}_{jd}$, and r_{jd} , then the chance constraints can be converted into

$$\Phi_{r_{jd}}^{-1}(\beta_{jd}) + \Phi_{s_{jd}}^{-1}(\beta_{jd}) + \Phi_{t_{j0d}}^{-1}(\beta_{jd}) \leq T, \quad \forall j \in N, d \in D.$$

Therefore, (M0) is equivalent to (M1).

4. Solution method

In this section, some basic concepts of multi-objective optimization are first presented. Then, the IMOABC algorithm is developed with improvements in the non-dominated solution set update strategy and neighborhood search techniques. Finally, the description of the LNS structure embedded in our proposed solution framework and the designed LNS operators are given.

4.1. Multi-objective optimization

The multi-objective optimization problems often have several conflicting objectives, usually, no perfect solution that achieves the best performance in each objective exists. Solutions in the context of multi-objective optimization are compared by Pareto dominance. For a multi-objective minimization problem with M independent objectives. Pareto dominance is defined as follows: solution x_1 dominates solution x_2 , denoted as $x_1 > x_2$ if and only if

$$\forall i = \{1, 2, \dots, M\} : f_i(x_1) \leq f_i(x_2) \quad \text{and} \\ \exists j = \{1, 2, \dots, M\} : f_j(x_1) < f_j(x_2).$$

If there is no other solution x in the decision space that dominates solution x' , then x' is the Pareto optimal solution. All Pareto optimal solutions constitute the Pareto optimal set. The set defined by the points mapped from the Pareto optimal set is called the Pareto frontier. In contrast to a weighted objective function, Pareto-based multi-objective optimization can find a set of non-dominated solutions that are equally good for the decision-maker.

4.2. Proposed IMOABC

Obviously, the proposed MoConHHRSP is NP-hard and more complicated than the previous ones due to its attribute of service consistency. Therefore, we develop a metaheuristic algorithm to obtain a set of non-dominated solutions to approximate the Pareto frontier. The designed IMOABC embeds the large neighborhood search heuristic and an adapted non-dominate solution set update strategy into the MOABC framework to improve the convergence and diversity of the MOABC algorithm.

Our proposed IMOABC is based on the MOABC framework (Akbari, Hedayatzadeh, Ziarati, & Hassanizadeh, 2012). The advantages of MOABC are few parameters, easy implementation, and simple calculation. However, two limitations of MOABC can be observed. First, in the population update, MOABC selects the better solutions by the Pareto dominance principle, however, in many cases, the new and old solutions may not dominate each other. Thus, MOABC ignores this situation and may miss some non-dominated solutions. In this way, the diversity of solutions may be reduced. In addition, MOABC searches solutions randomly to improve at each iteration, which may lead to low search efficiency of solutions and slow convergence. Based on these observations, we propose the IMOABC framework to improve the performance of the algorithm.

The IMOABC algorithm uses an adapted non-dominated solution set update strategy to diversify the obtained solution set. The non-dominated solution set update strategy is defined as follows. x' is the new solution searched in the neighborhood of solution x . If x is dominated by x' , it will be replaced. If the situation is the opposite, x' will be ignored. However, if x and x' are not dominated by each other, x' will be added to a supplementary archive A . After each iteration, the solutions stored in archive A will be merged with the new population and then the non-dominated solutions sorting is conducted to obtain the non-dominated solution set. In addition, the next population are generated from the non-dominated solutions set by the crowding distance of each solution in it. If the size of the non-dominated solutions set is less than the number of food sources, the remaining solutions are chosen from the current population by their

crowding distance. Another feature of the IMOABC solution framework is the use of the LNS heuristic. The efficiency of neighborhood search directly affects the solution quality. In order to improve the algorithm performance, the structure of the LNS heuristic is applied. By destroying and repairing feasible solutions in each iteration, a better solution can be obtained. A more detailed description of the applied LNS is provided in 4.3. Furthermore, multiple neighborhood operators that operate on different search spaces are utilized and explore the solution space in terms of the routing costs, caregiver consistency, and workload balance.

The overview of the developed IMOABC algorithm is provided in Algorithm 1. The bee colony consists of three types of bees: Employed, Onlooker, and Scout bees. The number of Employed bees and Onlooker bees is equal, and each Employed bee corresponds to a food source, thus the number of food sources is half of the population. The food sources represent the solutions to the optimization problem. In the initialization phase, the number of food sources H , the maximum trial value $MaxTrial$ and the number of iterations $MaxIter$ are input. H solutions are generated randomly as the initial solutions. Then the employed bees search new solutions in the neighborhoods using the LNS heuristic with a variety of local search operators. The solutions are updated following the defined update strategy. If one solution is not improved, then its $Trial$ value will plus one. Next, each onlooker bee chooses to follow one employed bee according to the roulette wheel principle and then explore a new food source in the neighborhood using the LNS heuristic. Solutions with higher quality will be selected and updated again in this phase. The probability of being selected is calculated by the formula

$$p_i = \frac{fit_i}{\sum_{i=1}^H fit_i},$$

where fit_i is the quality of solution i . In MOABC, fit_i is defined by the following formula:

$$fit_i = \frac{dom_i}{H},$$

where dom_i denotes the number of solutions dominated by solution i . The solutions are updated following the defined update strategy. If one solution is not improved, then its $Trial$ value will plus one. To avoid getting a local optimal solution, if there is no improvement for $MaxTrial$ times for a solution, i.e., the $Trial_i$ surpasses the $MaxTrial$ value, the employed bee will turn to a scout bee and initialize the solution randomly, the solution will be replaced by a regenerated initial solution. Finally, sort non-dominated solutions and update non-dominated solution sets.

The crowding distance measures the density of solutions. Choosing the solutions with a large crowding distance can guarantee the diversity of the solution set. It is defined as the average distance between two neighboring solutions on both sides of this solution according to each objective function. The solution set is first sorted according to the ascending order of the value of each objective function. For each objective function, the crowding distance of the first and last individual is set to infinity. All other intermediate solutions are set to the absolute difference of the normalized function values of two adjacent solutions. The crowded solutions with low crowding distance would perform badly in the diversity of solutions. As a sequence, when we select solutions for the next iteration, solutions with larger crowding distances are preferred and the solutions with low crowding distances would be discarded.

The initial solution is constructed by applying the greedy repair method. Using this method, customers are inserted iteratively at their best position until all customers are routed. In the construction of the initial solution, we only try to design the routes with regard to the total cost. Thus the initial solution may perform badly in the other objectives. However, we will improve the caregiver consistency and workload balance in the subsequent destroy and repair phase. The concrete operation of greedy repair can refer to Section 4.3.2.

Algorithm 1 IMOABC

Input: initial solution set N , the maximum trial value $MaxTrial$ and the number of iterations $MaxIter$

Output: non-dominated solution set F

```

1: Sort non-dominated solutions and obtain non-dominated solution set  $F$ ;
2: repeat
3:   a supplementary archive  $A$ 
4:   // employed bees' phase;
5:   for  $x$  in  $N$  do
6:     generate a new solution  $x'$  using LNS;
7:     update solution set  $N$  and supplementary archive  $A$  by the proposed non-dominate set update strategy
8:   end for
9:   calculate the selection possibility of each solution;
10:  // onlooker bees' phase;
11:  for  $m$  in  $N$  do
12:    select a solution  $x$  according to the roulette wheel principle and generate a new solution  $x'$  by LNS;
13:    update solution set  $N$  and supplementary archive  $A$  by the proposed non-dominate set update strategy;
14:  end for
15:  // scout bees' phase;
16:  for  $x$  in  $N$  do
17:    if  $I_x > Trail$  then
18:      randomly generate a new solution  $x'$  and replace  $x$ ;
19:    end if
20:    merge  $N$ ,  $F$ , and  $A$ , sort non-dominated solutions, and generate new population  $N$ ;
21:    update  $F$  by crowding distance;
22:  end for
23: until the stopping criterion is met

```

4.3. LNS

In this section, we first explain the large neighborhood search (LNS) framework used in the IMOABC. Then we describe the destroy and repair operators designed for the proposed problem.

Algorithm 2 presents the outline of the used LNS heuristic. The operation of each iteration is as follows: randomly select a destroy operator d to destroy the initial solution R . Daily routes are destroyed at the same time, once the customers that need to be removed are determined, all of them are removed from daily routes, and the destroyed solution is obtained. Then the repair operator is used to repair daily routes. In the repair operation, daily routes are repaired sequentially. One repair operator r is randomly selected to reinsert the removed customers into the route on a respective day. After all the daily routes are repaired, a new solution R' is generated.

Algorithm 2 LNS

Input: an initial daily routes R

Output: the final daily routes R'

```

select one destroy operator  $d$  randomly ;
 $R' = \text{destroy}(R, d)$ ;
for  $d \in D$  do
  select one repair operator  $r$  randomly ;
   $R'_d = \text{repair}(R'_d, s)$ ;
end for
merge  $R'_d$  and obtain the new daily routes  $R'$ ;
return  $R'$ 

```

4.3.1. Destroy operators

Destroy operators are utilized to remove customers from the current solution. The LNS used in our paper adopts three different removal

heuristics: Random removal, Related removal, and Worst caregiver consistency removal. The first step is to determine the number of the removed customers. We follow the method provided in [Ropke and Pisinger \(2006\)](#) to determine the number of removals. The number is a random number in the interval $\left[\min \left\{0.1 \frac{\sum_{d \in D} \sum_{n \in N} w_{id}}{|D|}, 30\right\}, \min \left\{0.4 \frac{\sum_{d \in D} \sum_{n \in N} w_{id}}{|D|}, 60\right\}\right]$. After the number of removed customers is determined, a destroy operator is adopted randomly to remove customers from the routes in the whole planning horizon.

Random removal. The random removal is the simplest removal heuristic which chooses σ customers randomly and deletes them from the current solution. This heuristic can greatly increase the diversity of local search.

Related removal. The related removal operator selects customers according to the relatedness between them. Generally, customers who are related are easier to interchange. The definition of the relatedness R_{ij} between customer i and customer j generally relies on the corresponding problem context. Therefore, in our problem, the relatedness R_{ij} consists of several perspectives, including the distance, the similarity of service time, the difference of service frequency. The weight of each component is measured by the parameters λ , μ , and ν .

$$R_{ij} = \lambda d_{ij} + \mu |\bar{s}_i - \bar{s}_j| + \nu \chi_{ij}.$$

where $\bar{s}_i$ and $\bar{s}_j$ is the mean service time of customer in multiple days. As service times are uncertain variables, they are calculated by their inverse uncertainty distributions. χ_{ij} denotes the similarity of service frequency, which is measured by the days of service requirement synchronism.

$$\chi_{ij} = |D| - \sum_{d \in D} |w_{id} - w_{jd}|$$

Worst caregiver consistency removal. The worst caregiver consistency removal operator is utilized to improve caregiver consistency of the current solution. It repeatedly chooses customers with the worst caregiver consistency so that they can be reinserted into positions that contribute to better caregiver consistency. Let I_i denote the value of caregiver inconsistency of customer i , then we can get I_i by

$$I_i = \begin{cases} v_i + \frac{1}{\log_2 v_i} (-\sum_{k \in K} \frac{b_{ik}}{s_i} \log_2 \frac{b_{ik}}{s_i}) & v_i \neq 1 \\ 1 & v_i = 1. \end{cases}$$

Customers are associated with respective values of caregiver inconsistency I_i , and they are sorted in descending order by I_i . In each iteration, customers are removed sequentially according to their order until σ customers have been removed.

4.3.2. Repair operators

After the destroy operators remove some customers from the current solution, the repair operators are in need to reinsert the removed customers into routes. Since our problem involves multi-day routes, we repair daily routes sequentially. In this paper, we adopt two types of repair operators: greedy operator and regret operator. Greedy operator inserts customers into the routes from the perspective of optimizing the total cost. Regret operator can facilitate workload balance. To expand the diversity of repair operator, three different variants of repair operator are also utilized. On each day, a repair operator is applied randomly to repair the daily route.

Greedy operator The basic greedy repair follows the cost-least principle. The removed customers are inserted in their best position one after another until all the removed customers are reassigned. The best insertion position of each customer is decided by the cost incurred to the existing routes and its feasibility in the resulting routes. In our operation, we define the cost as the change of the sum of travel cost, service cost, and penalty of late arrival. The cost is randomized by adding a random number for the incurred cost to increase the degree of diversification. Generally speaking, let $\Delta f_{i,k}$ denote the change in the total cost of inserting customer i into the cheapest position in route k .

We change the insertion cost $\Delta f_{i,k}$ to $\Delta f_{i,k} + r$, where r is a randomly generated number in the interval $[-\eta r, \eta r]$ and η defines the scale of randomization. The feasibility is checked from the maximum working time constraint. And we set $\Delta f_{i,k} = \infty$ if there is no feasible position in route k . The customer i_* is inserted into route k_* by the following formula:

$$(i^*, k^*) := \arg \min_{i \in N, k \in K} \Delta f_{i,k}.$$

For customers who have no feasible insertion position in the current routes, we add a new route to the solution.

Regret operators The basic greedy heuristic often selects the easy insertions in the beginning and put the difficult requests to the late iterations, which often lead to unbalanced customer number assigned to each caregiver. Thus the regret heuristics take the loss resulting from postponing an insertion to later iterations into account. Here, the insertion position of each customer is decided by the maximum difference between his best insertion position and other positions. Let Δf_i^q represent the change in the total cost caused by inserting customer i into his best position in the q th cheapest route. In each iteration, customer i is selected and inserted in by the formula

$$i^* := \arg \max_{i \in N, k \in K} \left(\sum_{h=2}^q \Delta f_i^h - \Delta f_i^1 \right).$$

Parameter q denotes the number of routes considered in the regret heuristics. We set $q = \{2, 3, 4\}$ to get three variants of the regret operator. In addition, the insertion cost is also randomized by adding a random number each time when we calculate the change in the total cost when a customer is inserted into a position in one route.

5. Computational experiments

In this section, we report the results of computational experiments to validate the effectiveness of the proposed model and algorithm. The data sets are designed with different characteristics to examine the performance of the algorithms and are introduced in Section 5.1. Section 5.2 describes the performance evaluation metrics used to measure the quality of multi-objective solution methods. In Section 5.3, we conduct the comparative experiments by comparing the proposed IMOABC with the MOABC and NSGAI and the results show the superiority of our proposed algorithm. Note that NSGAI is popular and can be referred to [Deb, Pratap, Agarwal, and Meyarivan \(2002\)](#). Section 5.4 summarizes the trade-off analysis.

5.1. Data sets and experimental setup

To the best of our knowledge, none of the benchmark datasets in the existing literature is applicable to our problem. Thus, we generate new sets of instances for the proposed problem. The data sets are adapted from Solomon's benchmark ([Solomon, 1987](#)). In Solomon's datasets, there are three types of customer location distribution, where R denotes that in this type customer locations are randomly distributed, C represents the clustered customer locations and mixed customer locations are in RC . We use the three types of instances and generate different sizes of problem instances with 10, 25, and 50 customers in each type. Next, we follow the procedure proposed by [Groer et al. \(2009\)](#) to generate the problem instances with a five-day planning horizon. The detailed information for the instances is given as follows.

For each instance, the coordinates of customers derive from Solomon's data. The Euclidean distance between two locations is usually taken as their travel time in deterministic models ([Zhan et al., 2021](#)). In our instances, we assume that travel times follow uncertainty normal distributions. The Euclidean distance is regarded as the mean value between each pair of customers, and the standard deviation is the multiple of the mean value with a value of 0.2. The original service times of all customers are set the same in the benchmark datasets. To be more practical, we diversify the service time of each customer. The

Table 1
Parameter setting of IMOABC.

<i>H</i>	MOABC		Repair operators	Removal operators		
	<i>MaxIter</i>	<i>MaxTrial</i>	η	λ	μ	ν
25	500	50 <i>H</i>	0.1	6	4	10

mean value of service time is generated from a uniform distribution [80, 100] for type *C* and [10, 30] for *R* and *RC* instances, the standard deviation is set in the same way as the travel time. The specified service start time is defined as the close time of each customer's time window. Unit travel cost, service cost, and penalty cost are set to 1.0. Considering the characteristics of different types of customers, we set service frequency to 0.5, 0.7, and 0.9. Service frequency is an indicator to measure the variation of customer visits. A service frequency of 100 percent means that each customer asks to be served every day. In this case, we can achieve perfect service consistency by executing the same routing plan every day within the scope of the plan. As the service decreases, the variations of customer service requests are even greater. We refer to the generated data sets as *Dataset_{0.5}*, *Dataset_{0.7}*, and *Dataset_{0.9}*, each data set contains 9 instances with 10, 25, and 50 customer numbers with different geographical distributions of the locations.

NSGAI is popular and has been widely used to solve multi-objective problems. We implement NSGAI following the description in Deb et al. (2002). The population size in NSGAI is set to 50. The crossover probability and mutation probability are set to 1 and 0.2. The number of NSGAI iterations is 1000. To clearly compare our IMOABC algorithm with the original MOABC, the parameters of the two algorithms are set to the same. To be fair, the size of the bee colony is set the same as that of NSGAI. Although the larger number of iterations increases the efficiency of the metaheuristic, large values of iterations also lead to excessive computational time consumption. To this end, the number of IMOABC and MOABC iterations is set to 500. Our IMOABC integrates LNS into the MOABC framework, so the random parameter η in the repair operator and the weights in the related removal λ , μ , ν are set. Table 1 provides the parameters in the IMOABC.

5.2. Performance metrics of multi-objective algorithms

Since the performance of multi-objective optimization algorithms cannot be evaluated directly in the means of the obtained objective value, the corresponding metrics to compare the quality of algorithms are proposed in the literature. The quality of multi-objective heuristics is measured from two aspects: convergence and diversity (diversity is subdivided into ductility and uniformity). Unary performance metrics usually calculate a single value to evaluate a certain aspect of quality. Binary performance metrics compare a pair of solution sets to measure the quality of heuristics by pair. Usually, a single performance metric has drawbacks and heuristics are evaluated by multiple performance metrics. In this paper, we use three unary metrics and one binary metric to evaluate the performance of the proposed IMOABC.

Spacing metric. Spacing metric measures the standard deviation of the minimum distance from each solution to the other solutions. The smaller the spacing value, the more uniform the solution set (Schott, 1995). For a Pareto solution set *P*, the value of the spacing metric is defined as follows:

$$Spacing(P) = \sqrt{\frac{1}{|P|-1} \sum_{i=1}^{|P|} (\bar{d} - d_i)^2},$$

where $d_i = \min_j \{ \sum_{k=1}^m |f_k(i) - f_k(j)| \}$, that is d_i represents the minimum distance of solution *i* to its closest solution *j*, the distance between two solutions is measured by the sum of the difference of each objective value among *m* objective values. $\bar{d}$ is the average value of all d_i .

However, this metric only measures the uniformity of the solution set, without considering the spread.

Set coverage. Set coverage (SC) is a binary operator to measure the convergence performance of algorithms by comparing the relative coverage of the two solution sets (Zitzler, Thiele, Laumanns, Fonseca, & Da Fonseca, 2003). Larger values of coverage indicate better performance. For two solution sets A and B, SC(A,B) is the percentage of the number of solutions in B that are dominated by at least one solution in A:

$$SC(A, B) = \frac{|b \in B : \exists a \in A, a \succ b|}{|B|}.$$

SC(A, B) = 1 means that all solutions in B are dominated by at least one solution in A. SC(A, B) = 0 means that no solution in B is dominated by the solution in A. SC(A, B) is not necessarily equal to 1 - SC(B, A).

Inverted generational distance. Inverted Generational Distance (IGD) is a unary indicator valuing the distance between Pareto optimal front P^* and *P* (Zhang & Li, 2007). The smaller the IGD value, the better the overall performance of the algorithm. It is defined as the average value of the distance from each point *x* in the Pareto optimal front P^* to the nearest solution in *P*:

$$IGD = \frac{\sum_{x \in P^*} d_x}{|P^*|},$$

where d_x represents the closest Euclidean distance between each point in P^* and *P*. As the real Pareto front is unknown, this paper constructs the approximate Pareto optimal front P^* for each instance by combining the non-dominated solutions obtained by all algorithms mentioned in this paper over three runs and then performing a non-dominated sorting to remove the dominated solutions.

Hypervolume. Hypervolume (HV) is a comprehensive unary indicator to assess the diversity and convergence of a non-dominated solution set (Coello, Lamont, Van Veldhuizen, et al., 2007). The larger the HV value, the better the performance of the algorithm. HV value is defined as the volume of the region in the target space covered by a non-dominated solution set *P*:

$$HV = \{ \cup_{x \in P} v_x \},$$

where v_x is the space covered by each point *x* ∈ *P* and a reference point *r*. The choice of the reference is important as it may affect the accuracy of this measure. In our proposed problem, v_x is defined as the objective space enclosed by the coordinates ($f_1(x)$, $f_2(x)$, $f_3(x)$) and the reference point *r*. As the three objectives considered in this paper do not have the same scale, it is necessary to normalize the objective values before computing the hypervolume metric. Thus we normalize the objective values based on the best and worst values for each objective obtained by all algorithms used in this paper over three runs. Then all objective values are normalized into the range [0, 1] and the reference point is set to (1.1, 1.1, 1.1) for all instances.

5.3. Performance comparison

We solve the test instances in three datasets by the described algorithms and report the performance metrics to evaluate the quality of the IMOABC. The non-dominated solution set of each instance is obtained by performing three independent runs. A non-dominated solution set is obtained for each run and then we merge the solution sets over the three runs. By conducting the non-dominated sorting on the aggregated solution sets, we obtain the non-dominated solution set for each instance.

Table 2, Table 3, and Table 4 provide the results of spacing, IGD, and HV metrics of each metaheuristic for the instances in *Dataset_{0.5}*, *Dataset_{0.7}*, and *Dataset_{0.9}*, respectively. The first column gives the instance number. The remaining columns show the three unary performance metrics of each metaheuristic for each instance. Each value in the table is the final result of the instance over three runs by

Table 2
Spacing, IGD and HV metrics results for *Dataset_{0.5}*.

Instance	Spacing			IGD			HV		
	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI
C10	0	0	0	0	0	0	NaN	NaN	NaN
R10	69.1823	6.2323*	13.5606	6.3460*	16.1494	7.6085	0.9361*	0.7533	0.8362
RC10	0*	12.9996	61.9221	0*	35.5273	35.5273	1.0000*	0.4452	0.5511
C25	89.3220*	659.7457	770.2111	0*	1687.2	487.4359	0.9428*	0.0998	0.1734
R25	35.0040	158.9077	27.8224*	0*	340.7316	111.1310	0.9061*	0.4157	0.5076
RC25	254.3428	56.2887	29.7235*	1.3817*	166.2190	42.2073	0.9119*	0.3026	0.7437
C50	905.2200	7405.7	507.4911*	0*	17682	7331.0	0.8350*	0.3892	0.3097
R50	126.0565*	128.8860	164.6413	0*	4852.7	3250.7	0.8349*	0.4125	0.5098
RC50	637.7075	103.1263	43.0502*	0*	2537.5	2027.0	0.9381*	0.5557	0.6158

Table 3
Spacing, IGD and HV metrics results for *Dataset_{0.7}*.

Instance	Spacing			IGD			HV		
	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI
C10	0	0	0	0	0	0	NaN	NaN	NaN
R10	13.9536*	122.6764	19.6556	1.7625*	88.9712	17.3511	0.9105*	0.6102	0.7721
RC10	8.4360*	159.1796	74.3611	0.4354*	75.2402	66.7827	0.8731*	0.7257	0.7648
C25	64.3546*	2779.4	510.3574	0*	7860.7	4735.3	0.9790*	0.5078	0.5921
R25	77.5787	106.1284	59.7934*	0*	1041.4	439.1046	0.8838*	0.3714	0.5464
RC25	42.8440	47.2458	35.1854*	0*	554.0027	294.8658	0.9179*	0.2354	0.4357
C50	1549.6	1862.8	753.9804*	0*	31252	18817	0.8328*	0.3736	0.3411
R50	371.0184	97.4046*	113.5492	0*	9097.8	6908.0	0.7673*	0.4266	0.4266
RC50	653.6522	69.8912*	142.4051	0*	6.4466	4661.1	0.8723*	0.4615	0.4871

Table 4
Spacing, IGD and HV metrics results for *Dataset_{0.9}*.

Instance	Spacing			IGD			HV		
	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI	IMOABC	MOABC	NSGAI
C10	0	0	0	0	0	0	NaN	NaN	NaN
R10	5.9018*	17.6748	22.6910	0*	352.3752	198.6398	0.9936*	0.5661	0.7365
RC10	10.3656*	28.9307	49.0311	17.2072*	429.0121	164.3708	0.9532*	0.2675	0.6685
C25	2.6415*	1433.5	289.6930	0*	14700	7631.1	0.9150*	0.2439	0.4107
R25	328.3720	60.5393	50.1514*	0*	2.1559	1.2370	0.9150*	0.4646	0.5990
RC25	207.7884	595.0274	44.0101*	0*	2065.5	1341.7	0.9324*	0.4257	0.5067
C50	2785.5	1347.5	594.3909*	21.4685*	4591.2	3131.6	0.8013*	0.4250	0.5442
R50	220.5615	249.2830	130.4885*	0*	14543	13110	0.7417*	0.3758	0.4267
RC50	660.1384	485.1596	81.9130*	0*	12064	11292	0.8378*	0.4342	0.5408

the corresponding algorithm. The best results of each metric for each instance are marked with *.

As indicated in these tables, IMOABC strongly outperforms the other algorithms in terms of IGD and HV metrics, which validates that our IMOABC achieves the best overall performance than MOABC and NSGAI. Especially, for the instance C10, we can see that IGD values of all the three algorithms are 0 and HV values are NaN. The reason is that there is only one solution obtained for this problem instance no matter which algorithm is used. Further explanation is that due to the small number of customers and clustered geographical distributions of customers, the route plans can always achieve best caregiver consistency, i.e., a caregiver is assigned to service the small set of clustered customers over the planning horizon, thus workload balance also gets the perfect value. Based on the best values of caregiver consistency and workload balance, the algorithms will always try to obtain the optimal solution that performs best on the objective of minimizing total costs. We also observe that the IGD value of the IMOABC is 0 in most instances in three datasets, which represents that the Pareto optimal front obtained by the three algorithms for the corresponding instance is the solution set obtained by IMOABC. The solutions generated by the proposed IMOABC cannot be dominated by the other solutions obtained by the other algorithms, which further validates the superiority of the IMOABC in the quality of solutions. However, our IMOABC performs worse in spacing metric than NSGAI, for example, in Table 2, IMOABC is outperformed by NSGAI on 5 out of 9 instances, while IMOABC outperforms MOABC on 3 instances compared to 1 instance of MOABC. In general, our IMOABC strongly outperforms

Table 5
SC metric results.

Dataset	SC	MOABC	NSGAI
<i>Dataset_{0.5}</i>	SC(*,IMOABC)	0.00%	1.94%
	SC(IMOABC,*)	66.67%	74.28%
<i>Dataset_{0.7}</i>	SC(*,IMOABC)	0.00%	2.31%
	SC(IMOABC,*)	65.08%	64.81%
<i>Dataset_{0.9}</i>	SC(*,IMOABC)	0.00%	0.00%
	SC(IMOABC,*)	66.67%	65.74%

MOABC in spacing, IGD, and HV metrics and weakly outperforms NSGAI. Finally, Fig. 2(a)–(c) gives an illustrative presentation of the Pareto solutions obtained by the algorithms used in this paper for instances with 25 customers in *Dataset_{0.5}*.

The comparisons of the set coverage metrics among the algorithms for three datasets are summarized in Table 5. The first column is the dataset. SC(*, IMOABC) value indicates the percentage of solutions obtained by IMOABC dominated by at least one solution generated by the comparison algorithm *. The value is the average of set coverage of 9 instances in each dataset. From Table 5, it can be easily observed that MOABC cannot generate solutions that dominate the solutions found by the IMOABC, with the value of 0.00% in three datasets. While, the IMOABC can always achieve solutions that dominate those obtained in MOABC, with the value of at least 65.08% in the set coverage metric. NSGAI can find some solutions that are not dominated by our proposed IMOABC, however, the percentage is very small, with only

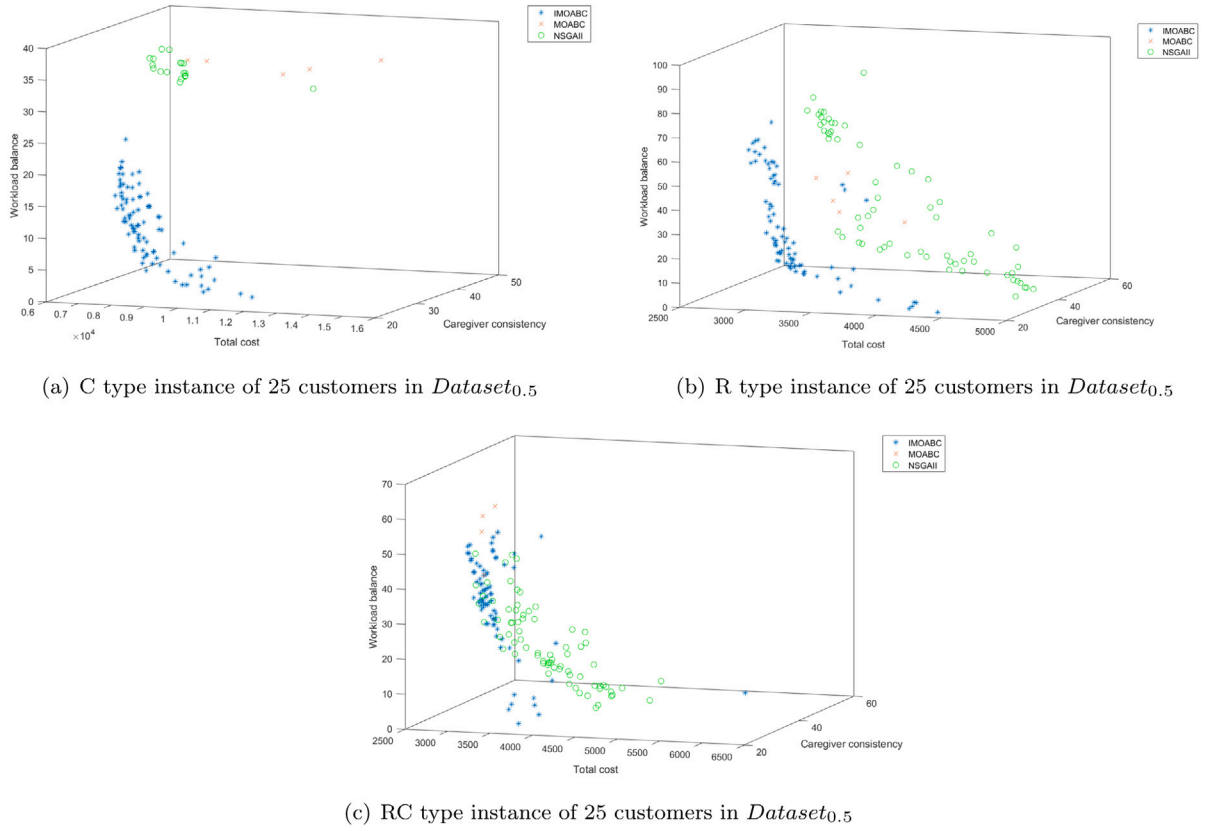


Fig. 2. Example non-dominated solution set obtained by the algorithms.

1.94% and 2.31% in *Dataset*_{0.5} and *Dataset*_{0.7}, respectively. Overall, the proposed IMOABC can find the best non-dominated solution set among the comparison algorithms for all datasets.

5.4. Trade-off analysis

As stated before, the result found by the proposed multi-objective algorithms is a set of non-dominated solutions. For the decision-makers, too many solutions will also be confusing, the best trade-off solution is more desirable to apply in practice. Therefore, in this section, we conduct the trade-off analysis on the Pareto solution set obtained by the proposed IMOABC over three runs. To analyze the trade-off relationship among multiple objectives, we adopt the level diagram technique introduced by [Blasco, Herrero, Sanchis, and Martínez \(2008\)](#). Since our objectives are not the same scale, we first normalize the objective values of non-dominated solutions into the range of [0, 1]. Then for each solution s , we can get a value by the formula in [Blasco et al. \(2008\)](#). Note that the smaller the value, the better the compromise solution.

Confronted with a set of non-dominated solutions, the decision-maker can decide the best one based on his decision criterion, fully focusing on one objective among the three objectives or choosing the compromise solution. Therefore, we present the solutions for each instance under the four decision criteria. Information on solutions based on the four decision criterion for each instance is provided in [Table 6](#). s^{BC} denotes the best compromise solution in the non-dominated solution set. Furthermore, the extreme solutions in the non-dominated solution set, the one with the best travel cost, the one with the best caregiver consistency, and the one with the best workload balance are denoted by s^{TC} , s^{CC} , and s^{WB} . For each instance, the non-dominated solution set is the combined results over three runs after removing the dominated solution. The first two columns give the dataset and instance name. The remaining columns show the values of the three

objectives under four decision criteria. In general, we can find that there exists a significant difference among the four solutions in terms of three objective values. As a consequence, the decision criteria on which the service providers bases to make decisions impacts considerably on operating costs, caregiver consistency, and workload balance. Take the instance R25 of *Dataset*_{0.5} as an example, we focus on analyzing the best compromise solution s^{BC} for this instance in [Table 6](#). First, we can observe that if we put all efforts into the optimization of the total cost, the best total cost in s^{TC} for this instance is 2795.31, while the caregiver consistency and workload balance values are 40.57 and 52, respectively. Comparing with the best compromise solution s^{BC} , we can find that the total cost is 3125.28, approximately 11.8% higher than that in the best total cost solution s^{TC} . However, the increase in travel cost contributes to the improvements in caregiver consistency and workload balance, with caregiver consistency of 27 (a 33.45% decrease) and 21 (a 59.62% decrease), respectively.

Summary statistics of the trade-offs among the three objectives are reported in [Table 7](#). In [Table 7](#), we compare the best compromise solution with the three extreme solutions in each objective: the best travel cost solution, the best caregiver consistency solution, and the best workload balance solution. The values in this table are the average values of the improvement or deterioration on each objective value compared with the best compromise solution of all the instances. The first row indicates that if the decision-makers adopt the best compromise solution, they will sacrifice total cost by 18.41%, while great improvements in workload balance and slight improvements in caregiver consistency can be achieved, on average, when compared with the extreme solution with the best routing cost, which indicates that an approximate 60% improvement in caregiver consistency and 3.5 times improvement in workload balance come at the price of an increase of 18% in routing cost. When comparing the solution with the best caregiver consistency with the compromise solution, we can see that a slight improvement in caregiver consistency of 20.10% will lead

Table 6
Solutions under different decision criteria.

Dataset	instance	s^{BC}			s^{TC}			s^{CC}			s^{WB}		
		f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3	f_1	f_2	f_3
Dataset _{0.5}	C10	2724.89	10	0	2724.89	10	0	2724.89	10	0	2724.89	10	0
	R10	1565.27	11.92	1	1561.30	13.92	1	1654.80	10	1	1654.80	10	1
	RC10	1494.74	10	1	1494.74	10	1	1494.74	10	1	1494.74	10	1
	C25	8061.57	26.92	5	6716.50	36.89	15	10529.14	25	7	8937.23	30.65	1
	R25	3125.28	27.00	21	2795.31	40.57	52	3832.18	25	45	4375.74	25	0
	RC25	3463.55	27.95	4	2881.94	31.92	49	6351.60	25	13	3576.88	35.42	0
	C50	35001.04	63.18	18	14670.72	112.87	69	93214.20	50	49	30575.93	96.48	2
	R50	8193.86	75.12	82	6276.48	109.69	95	26802.04	50	211	6983.78	95.83	6
Dataset _{0.7}	RC50	8651.84	65.32	62	6757.01	93.16	153	47745.49	50	208	9681.85	78.72	22
	C10	3569.38	10	0	3569.38	10	0	3569.38	10	0	3569.38	10	0
	R10	1971.50	15.54	16	1954.74	17.26	18	2206.35	10	1	2206.35	10	1
	RC10	1983.50	13.72	1	1851.62	21.65	26	2104.24	10	1	1983.50	13.72	1
	C25	10768.56	26.81	2	9438.03	37.51	62	11132.48	25	0	10770.50	32.62	0
	R25	5171.78	39.72	18	4393.30	63.51	72	9081.29	25	68	4778.34	49.06	6
	RC25	4326.51	36.67	18	4184.42	54.37	46	7112.93	25	32	6138.82	28.53	4
	C50	56930.09	83.47	41	22139.53	153.46	106	226134.71	53.44	238	44328.84	111.82	11
Dataset _{0.9}	R50	13647.09	101.83	196	9054.35	157.00	326	62974.71	55.64	500	11691.29	134.48	28
	RC50	17092.71	78.99	171	9161.58	116.40	215	60631.20	51.92	466	9322.87	128.37	39
	C10	5243.35	10	0	5243.35	10	0	5243.35	10	0	5243.35	10	0
	R10	2689.23	11.97	2	2627.98	23.58	4	3341.19	10	6	2636.46	15.17	2
	RC10	2436.81	13.44	2	2349.54	17.39	6	2378.61	10	6	2730.05	11.72	1
	C25	12577.36	32.23	6	12427.02	39.84	16	62288.44	25	17	12747.37	38.34	2
	R25	8637.00	40.37	13	6262.81	72.77	34	25947.12	28.44	175	7283.74	72.35	5
	RC25	5931.57	32.14	13	5756.02	36.64	13	14519.05	25	103	5803.74	41.36	5
Dataset _{0.9}	C50	57144.03	101.95	24	28498.74	163.61	215	213511.01	55.26	192	32973.11	163.93	6
	R50	21636.93	117.19	204	11275.28	172.56	93	86840.14	57.69	948	13739.85	177.06	40
	RC50	21457.61	92.36	277	11917.32	147.89	223	87891.33	50	669	13034.06	124.97	110

Table 7
Statistics of comparison results of four solutions.

	f_1	f_2	f_3
s^{TC} vs s^{BC}	18.41% ↗	39.64% ↘	353.63% ↘
s^{CC} vs s^{BC}	141.33% ↘	20.10% ↗	297.33% ↘
s^{WB} vs s^{BC}	0.08% ↗	22.11% ↘	57.67% ↗

to significant deteriorations in the other objectives, an approximate 1.5 times decrease in routing cost and triple drop in workload balance. Another interesting finding in this table is that when the manager totally focuses on the objective of workload balance, great improvement on this objective can be achieved with slight negative impacts on caregiver consistency. In particular, the effect on the total cost is slightly positive.

6. Conclusions

In this paper, we introduce a multi-objective home healthcare routing and scheduling problem taking into consideration multiple realistic factors such as working time regulation, specified service start time, service consistency, and workload balance. In addition to the traditional objective of cost optimization, customer satisfaction and caregiver satisfaction receive equal attention with caregiver consistency and workload balance modeled as independent optimization objectives. Due to the uncertainties in the operation of the home care industry, uncertain travel times and service times are also considered in the formulation of the MoConHHRSP model. Uncertainty theory is applied for the first time to deal with the uncertain variables and transform the uncertain programming model into crisp equivalents. To solve the proposed model, the IMOABC algorithm is designed which improves the non-dominate solution update strategy and utilizes a large neighborhood search (LNS) to conduct a local search. The performance of IMOABC is compared with popular NSGAI and MOABC from four metrics: spacing, IGD, SC, and HV. A series of computational experiments demonstrate the efficiency of the proposed IMOABC. In addition, the trade-off analysis is also conducted. The findings are summarized as follows:

(1) The developed IMOABC outperforms NSGAI and MOABC in evaluation metrics IGD, SC, and HV, which validates the superiority of IMOABC in both convergence and diversity of the obtained non-dominated solution set.

(2) The trade-off analysis reveals that a better caregiver consistency can be achieved at a high price of total costs and workload balance, while a great improvement on the workload balance can be provided with little deterioration in caregiver consistency.

(3) In many cases, low total cost and high level of workload balance can be achieved simultaneously.

Some extensions for this paper can be further studied from the following aspects. The home health care routing and scheduling problem is complicated involving multiple decision optimization, thus the introduction of practical factors can be an interesting direction for further work. For instance, the factors such as green emissions and the skill requirement of caregivers can be incorporated. The second direction of improvement is the efficiency of the solution algorithm, other strategies can be added to the algorithm to accelerate the speed of the algorithm. Furthermore, the dynamic variants of this problem is also a promising extension in the future study.

CRediT authorship contribution statement

Meng Yang: Conceptualization, Methodology, Software, Investigation, Writing - original draft, Writing - review & editing. **Yaodong Ni:** Conceptualization, Methodology, Writing - review & editing, Funding acquisition, Supervision. **Liu Yang:** Writing - review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 71571043, 71471038), Program for Huiyuan Distinguished Young Scholars, China, UIBE (No. 17JQ09, 17JQ06), and “The Fundamental Research Funds for the Central Universities, China” in UIBE (No. CXTD10-05, CXTD11-04).

Appendix

Here some basic concepts and theorems of uncertainty theory are introduced.

Let Γ be a nonempty set, and $\mathcal{L}$ be a σ -algebra over Γ . Each element $A \in \mathcal{L}$ is called an event. A number $\mathcal{M}\{A\}$ indicates the possibility that A will occur. Uncertain measure $\mathcal{M}$ is introduced as a set function satisfying the following axioms (Liu, 2007):

Axiom 1 (Normality Axiom). $\mathcal{M}\{\Gamma\} = 1$ for the universal set Γ .

Axiom 2 (Duality Axiom). $\mathcal{M}\{A\} + \mathcal{M}\{A^c\} = 1$ for any event A .

Axiom 3 (Subadditivity Axiom). For every countable sequence of events $\{A_i\}$, we have

$$\mathcal{M}\left\{\bigcup_{i=1}^{\infty} A_i\right\} \leq \sum_{i=1}^{\infty} \mathcal{M}\{A_i\}.$$

The triplet $(\Gamma, \mathcal{L}, \mathcal{M})$ is called an uncertainty space. In addition, the product uncertain measure (Liu, 2009) was defined as following.

Axiom 4 (Product Axiom). Let $(\Gamma_k, \mathcal{L}_k, \mathcal{M}_k)$ be uncertainty spaces for $k = 1, 2, \dots$. The product uncertain measure $\mathcal{M}$ is an uncertain measure satisfying

$$\mathcal{M}\left\{\prod_{k=1}^{\infty} A_k\right\} = \bigwedge_{k=1}^{\infty} \mathcal{M}_k\{A_k\}.$$

Definition 1 (Liu, 2007). An uncertain variables is a function ξ from an uncertainty space $(\Gamma, \mathcal{L}, \mathcal{M})$ to the set of real numbers such that $\xi \in B$ is an event for any Borel set B .

Definition 2 (Liu, 2007). The uncertainty distribution Φ of an uncertain variable ξ is defined by

$$\Phi(x) = \mathcal{M}\{\xi \leq x\}, \forall x \in \Re$$

for any real number x .

Definition 3 (Liu, 2010). An uncertainty distribution $\Phi(x)$ is said to be regular if it is a continuous and strictly increasing function with respect to x at which $0 < \Phi(x) < 1$, and

$$\lim_{x \rightarrow -\infty} \Phi(x) = 0, \quad \lim_{x \rightarrow +\infty} \Phi(x) = 1.$$

Definition 4 (Liu, 2007). Let ξ be an uncertain variable. Then the expected value of ξ is defined by

$$E[\xi] = \int_0^{+\infty} \mathcal{M}\{\xi \geq r\} dr - \int_{-\infty}^0 \mathcal{M}\{\xi \leq r\} dr$$

provided that at least one of the two integrals is finite.

Theorem 2 (Liu, 2010). Let $\xi_1, \xi_2, \dots, \xi_n$ are independent uncertain variables with regular uncertainty distributions $\Phi_1, \Phi_2, \dots, \Phi_n$, respectively. If $f(\xi_1, \xi_2, \dots, \xi_n)$ is strictly increasing with respect to $\xi_1, \xi_2, \dots, \xi_m$ and strictly decreasing with respect to $\xi_{m+1}, \xi_{m+2}, \dots, \xi_n$, then

$$\xi = f(\xi_1, \xi_2, \dots, \xi_n)$$

has an inverse uncertainty distribution

$$\Psi^{-1}(\alpha) = f(\Phi_1^{-1}(\alpha), \dots, \Phi_m^{-1}(\alpha), \Phi_{m+1}^{-1}(1-\alpha), \dots, \Phi_n^{-1}(1-\alpha)).$$

Theorem 3 (Liu, 2010). Let ξ be an uncertain variable with regular uncertainty distribution Φ . Then

$$E[\xi] = \int_0^1 \Phi^{-1}(\alpha) d\alpha.$$

Theorem 4 (Liu, 2010). Assume $\xi_1, \xi_2, \dots, \xi_n$ are independent uncertain variables with regular uncertainty distributions $\Phi_1, \Phi_2, \dots, \Phi_n$, respectively. If $f(\xi_1, \xi_2, \dots, \xi_n)$ is strictly increasing with respect to $\xi_1, \xi_2, \dots, \xi_m$ and strictly decreasing with respect to $\xi_{m+1}, \xi_{m+2}, \dots, \xi_n$, then

$$\xi = f(\xi_1, \xi_2, \dots, \xi_n)$$

has an expected value

$$E[\xi] = \int_0^1 f(\Phi_1^{-1}(\alpha), \dots, \Phi_m^{-1}(\alpha), \Phi_{m+1}^{-1}(1-\alpha), \dots, \Phi_n^{-1}(1-\alpha)) d\alpha.$$

Theorem 5 (Liu, 2010). Assume the constraint function $g(x, \xi_1, \xi_2, \dots, \xi_n)$ is strictly increasing with respect to $\xi_1, \xi_2, \dots, \xi_m$ and strictly decreasing with respect to $\xi_{m+1}, \xi_{m+2}, \dots, \xi_n$. If $\xi_1, \xi_2, \dots, \xi_n$ are independent uncertain variables with uncertainty distributions $\Phi_1, \Phi_2, \dots, \Phi_n$ respectively, then the chance constraint

$$\mathcal{M}\{g(x, \xi_1, \xi_2, \dots, \xi_n) \leq 0\} \geq \alpha$$

holds if and only if

$$g(x, \Phi_1^{-1}(\alpha), \dots, \Phi_m^{-1}(\alpha), \Phi_{m+1}^{-1}(1-\alpha), \dots, \Phi_n^{-1}(1-\alpha)) \leq 0.$$

References

- Akbari, R., Hedayatzadeh, R., Ziarati, K., & Hassanizadeh, B. (2012). A multi-objective artificial bee colony algorithm. *Swarm and Evolutionary Computation*, 2, 39–52.
- Akjaritakarl, C., Yenradee, P., & Drake, P. R. (2007). PSO-based algorithm for home care worker scheduling in the UK. *Computers & Industrial Engineering*, 53(4), 559–583.
- Alodhayani, A. A. (2017). Comparison between home health care and hospital services in elder population: cost-effectiveness.
- Begur, S. V., Miller, D. M., & Weaver, J. R. (1997). An integrated spatial DSS for scheduling and routing home-health-care nurses. *Interfaces*, 27(4), 35–48.
- Blasco, X., Herrero, J. M., Sanchis, J., & Martínez, M. (2008). A new graphical visualization of n-dimensional Pareto front for decision-making in multiobjective optimization. *Information Sciences*, 178(20), 3908–3924.
- Braekers, K., Hartl, R. F., Parragh, S. N., & Tricoire, F. (2016). A bi-objective home care scheduling problem: Analyzing the trade-off between costs and client inconvenience. *European Journal of Operational Research*, 248(2), 428–443.
- Braekers, K., & Kovacs, A. A. (2016). A multi-period dial-a-ride problem with driver consistency. *Transportation Research, Part B (Methodological)*, 94, 355–377.
- Campelo, P., Nevesmoreira, F., Amorim, P., & Almadalobo, B. (2019). Consistent vehicle routing problem with service level agreements: A case study in the pharmaceutical distribution sector. *European Journal of Operational Research*, 273(1), 131–145.
- Cissé, M., Yalçındağ, S., Kergosien, Y., Şahin, E., Lenté, C., & Matta, A. (2017). OR problems related to home health care: A review of relevant routing and scheduling problems. *Operations Research for Health Care*, 13, 1–22.
- Coello, C. A. C., Lamont, G. B., Van Veldhuizen, D. A., et al. (2007). *Evolutionary algorithms for solving multi-objective problems* (vol. 5). Springer.
- Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2), 182–197.
- Decerle, J., Grunder, O., El Hassani, A. H., & Barakat, O. (2019). A hybrid memetic-ant colony optimization algorithm for the home health care problem with time window, synchronization and working time balancing. *Swarm and Evolutionary Computation*, 46, 171–183.
- Duque, P. M., Castro, M., Sörensen, K., & Goos, P. (2015). Home care service planning. The case of Landelijke Thuiszorg. *European Journal of Operational Research*, 243(1), 292–301.
- Fathollahi-Fard, A. M., Ahmadi, A., Goodarzian, F., & Cheikhrouhou, N. (2020). A bi-objective home healthcare routing and scheduling problem considering patients' satisfaction in a fuzzy environment. *Applied Soft Computing*, Article 106385.
- Feillet, D., Garaix, T., Lehuède, F., Peton, O., & Quadri, D. (2014). A new consistent vehicle routing problem for the transportation of people with disabilities. *Networks*, 63(3), 211–224.
- Fikar, C., & Hirsch, P. (2017). Home health care routing and scheduling: A review. *Computers & Operations Research*, 77, 86–95.
- Genet, N., Boerma, W., Kroneman, M., Hutchinson, A., & Saltman, R. (2013). *Home care across Europe: case studies*. WHO.

- Goodarzi, A. H., Tavakkoli-Moghaddam, R., & Amini, A. (2020). A new bi-objective vehicle routing-scheduling problem with cross-docking: mathematical model and algorithms. *Computers & Industrial Engineering*, 149, Article 106832.
- Groer, C., Golden, B. L., & Wasil, E. (2009). The consistent vehicle routing problem. *Manufacturing and Service Operations Management*, 11(4), 630–643.
- Harris-Kojetin, L. D., Sengupta, M., Park-Lee, E., & Valverde, R. (2013). Long-term care services in the United States: 2013 overview.
- Janssens, J., Van den Bergh, J., Sörensen, K., & Catrysse, D. (2015). Multi-objective microzone-based vehicle routing for courier companies: From tactical to operational planning. *European Journal of Operational Research*, 242(1), 222–231.
- Kovacs, A. A., Golden, B. L., Hartl, R. F., & Parragh, S. N. (2015). The generalized consistent vehicle routing problem. *Transportation Science*, 49(4), 796–816.
- Kovacs, A. A., Parragh, S. N., & Hartl, R. F. (2014). A template-based adaptive large neighborhood search for the consistent vehicle routing problem. *Networks*, 63(1), 60–81.
- Kovacs, A. A., Parragh, S. N., & Hartl, R. F. (2015). The multi-objective generalized consistent vehicle routing problem. *European Journal of Operational Research*, 247(2), 441–458.
- Lian, K., Milburn, A. B., & Rardin, R. L. (2016). An improved multi-directional local search algorithm for the multi-objective consistent vehicle routing problem. *IIE Transactions*, 48(10), 975–992.
- Lin, C. C., Hung, L. P., Liu, W. Y., & Tsai, M. C. (2018). Jointly rostering, routing, and rerostering for home health care services: A harmony search approach with genetic, saturation, inheritance, and immigrant schemes. *Computers & Industrial Engineering*, 115, 151–166.
- Liu, B. (2007). *Uncertainty theory*. Springer-Verlag.
- Liu, B. (2009). Some research problems in uncertainty theory. *Journal of Uncertain Systems*, 3(1), 3–10.
- Liu, B. (2010). *Uncertainty theory: A branch of mathematics for modeling human uncertainty*. DBLP.
- Liu, R., Xie, X., & Garaix, T. (2013). Weekly home health care logistics. In *2013 10th IEEE International conference on networking, sensing and control* (pp. 282–287). IEEE.
- Liu, M., Yang, D., Su, Q., & Xu, L. (2018). Bi-objective approaches for home healthcare medical team planning and scheduling problem. *Computational & Applied Mathematics*, 37(4), 4443–4474.
- Liu, R., Yuan, B., & Jiang, Z. (2017). Mathematical model and exact algorithm for the home care worker scheduling and routing problem with lunch break requirements. *International Journal of Productions Research*, 55(2), 558–575.
- Liu, R., Yuan, B., & Jiang, Z. (2019). A branch-and-price algorithm for the home-caregiver scheduling and routing problem with stochastic travel and service times. *Flexible Services and Manufacturing Journal*, 31(4), 989–1011.
- Luo, Z., Qin, H., Che, C. H., & Lim, A. (2015). On service consistency in multi-period vehicle routing. *European Journal of Operational Research*, 243(3), 731–744.
- Matl, P., Hartl, R. F., & Vidal, T. (2018). Workload equity in vehicle routing problems: A survey and analysis. *Transportation Science*, 52(2), 239–260.
- Matl, P., Hartl, R. F., & Vidal, T. (2019). Workload equity in vehicle routing: The impact of alternative workload resources. *Computers & Operations Research*, 110, 116–129.
- Milburn, A. B., & Spicer, J. (2013). Multi-objective home health nurse routing with remote monitoring devices. *International Journal of Planning and Scheduling*, 1(4), 242–263.
- Paquette, J., Bellavance, F., Cordeau, J.-F., & Laporte, G. (2012). Measuring quality of service in dial-a-ride operations: the case of a Canadian city. *Transportation*, 39(3), 539–564.
- Rodriguezmartin, I., Salazar-gonzalez, J., & Yaman, H. (2019). The periodic vehicle routing problem with driver consistency. *European Journal of Operational Research*, 273(2), 575–584.
- Ropke, S., & Pisinger, D. (2006). An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation Science*, 40(4), 455–472.
- Russell, D., Rosati, R. J., Rosenfeld, P., & Marren, J. M. (2011). Continuity in home health care: is consistency in nursing personnel associated with better patient outcomes? *Journal for Healthcare Quality*, 33(6), 33–39.
- Sasser, W. E., Schlesinger, L. A., & Heskett, J. L. (1997). *Service profit chain*. Simon and Schuster.
- Schott, J. R. (1995). *Fault tolerant design using single and multicriteria genetic algorithm optimization* (Ph.D. thesis), Massachusetts Institute of Technology.
- Shi, Y., Boudouh, T., & Grunder, O. (2019). A robust optimization for a home health care routing and scheduling problem with consideration of uncertain travel and service times. *Transportation Research Part E: Logistics and Transportation Review*, 128, 52–95.
- Solomon, M. M. (1987). Algorithms for the vehicle routing and scheduling problems with time window constraints. *Operations Research*, 35(2), 254–265.
- Song, Y., Ulmer, M. W., Thomas, B. W., & Wallace, S. W. (2020). Building trust in home services—stochastic team-orienting with consistency constraints. *Transportation Science*, 54(3), 823–838.
- Spiliot, R., Dabia, S., & Van Woensel, T. (2018). The time window assignment vehicle routing problem with time-dependent travel times. *Transportation Science*, 52(2), 261–276.
- Spiliot, R., & Gabor, A. F. (2015). The time window assignment vehicle routing problem. *Transportation Science*, 49(4), 721–731.
- Tarantilis, C. D., Stavropoulou, F., & Repoussis, P. P. (2012). A template-based tabu search algorithm for the consistent vehicle routing problem. *Expert Systems with Applications*, 39(4), 4233–4239.
- Trautsumwieser, A., & Hirsch, P. (2014). A branch-price-and-cut approach for solving the medium-term home health care planning problem. *Networks*, 64(3), 143–159.
- Woodward, C. A., Abelson, J., Tedford, S., & Hutchison, B. (2004). What is important to continuity in home care?: Perspectives of key stakeholders. *Social Science and Medicine*, 58(1), 177–192.
- Xu, X., Hao, J., & Zheng, Y. (2020). Multi-objective artificial bee colony algorithm for multi-stage resource leveling problem in sharing logistics network. *Computers & Industrial Engineering*, 142, Article 106338.
- Yang, M., Ni, Y., & Song, Q. (2020). Optimizing driver consistency in the vehicle routing problem under uncertain environment, Working Paper, University of International Business and Economics, Beijing.
- Yuan, B., Liu, R., & Jiang, Z. (2015). A branch-and-price algorithm for the home health care scheduling and routing problem with stochastic service times and skill requirements. *International Journal of Productions Research*, 53(24), 7450–7464.
- Yuan, B., Liu, R., & Jiang, Z. (2018). Daily scheduling of caregivers with stochastic times. *International Journal of Productions Research*, 56(9), 3245–3261.
- Zhan, Y., & Wan, G. (2018). Vehicle routing and appointment scheduling with team assignment for home services. *Computers & Operations Research*, 100, 1–11.
- Zhan, Y., Wang, Z., & Wan, G. (2021). Home service routing and appointment scheduling with stochastic service times. *European Journal of Operational Research*, 288(1), 98–110.
- Zhang, Q., & Li, H. (2007). MOEA/D: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6), 712–731.
- Zhen, L., Lv, W., Wang, K., Ma, C., & Xu, Z. (2020). Consistent vehicle routing problem with simultaneous distribution and collection. *Journal of the Operational Research Society*, 71(5), 813–830.
- Zitzler, E., Thiele, L., Laumanns, M., Fonseca, C. M., & Da Fonseca, V. G. (2003). Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Transactions on Evolutionary Computation*, 7(2), 117–132.